Historical Prediction Attention Mechanism-Based Trajectory Forecasting for Proactive Work Zone Safety in a Digital Twin Environment

Minhaj Uddin Ahmad, Mizanur Rahman, *Senior Member, IEEE*, Alican Sevim, David Bodoh, Sakib Mahmud Khan, Li Zhao, Nathan Huynh, and Eren Erman Ozguven

***Abstract*—Proactive safety systems aim to mitigate risks by anticipating future potential conflicts between vehicles and enabling early intervention to prevent work-zone-related crashes. This study presents an infrastructure-enabled proactive work zone safety warning system that leverages a digital twin (DT) environment, integrating real-time multisensor data, detailed high-definition (HD) maps, and a trajectory prediction model based on a historical-prediction attention mechanism (AM). Specifically, the contributions of this study are in twofolds: 1) developing an infrastructure-enabled vehicle trajectory prediction framework that leverages the historical prediction network (HPNet) model integrated with Lanelet2 HD maps and 2) developing a proactive safety warning generation application that incorporates probabilistic conflict modeling and vehicle bounding-box representations. The development and effectiveness demonstration of the presented trajectory prediction model and proactive warning system is conducted using a cosimulation environment integrating simulation of urban mobility (SUMO) and CAR learning to act (CARLA) simulators. To evaluate the accuracy of predicted trajectories (PTs), we use two standard metrics: joint average displacement error (ADE) and joint final displacement error (FDE). Our analyses revealed that the infrastructure-enabled HPNet model demonstrates superior performance on the work zone datasets generated from the cosimulation environment, achieving a minimum joint FDE of 0.3228 m and a minimum joint ADE of 0.1327 m, lower than the benchmarks on the Argoverse (minimumJoint FDE: 1.0986 m, minimumJoint ADE: 0.7612 m) and INTERACTION (minimumJoint FDE: 0.8231 m, minimumJoint ADE: 0.2548 m) datasets. In addition, our proactive safety warning generation application demonstrates its ability to issue alerts for potential vehicle conflicts.**

***Index Terms*—Conflict prediction, digital twin (DT), high-definition (HD) maps, proactive safety, trajectory prediction, work zone safety.**

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I. Introduction

ROADWAY safety remains a critical issue, particularly within work zones where vehicle dynamics and driver behavior significantly vary due to altered roadway configurations and proximity to workers. Work zones pose unique risks as drivers navigate narrowed lanes, altered traffic patterns, and speed limit changes, often in close proximity to roadway workers. Work-zone-related crashes have consistently presented higher fatality rates compared with other roadway segments [1]. A 2022 Bureau of Transportation Statistics report shows a 14-year-high crash record of 856 fatal crashes in work zones [2]. A 2022 nationwide study on highway work zone safety by the Associated General Contractors of America (AGC) found that two-thirds of highway contractors have encountered work zone crashes. In addition, over one-third (38%) reported project delays due to crashes, and 5% reported worker fatalities [3]. To address these persistent challenges, proactive safety strategies have emerged as a promising solution. A proactive system that anticipates vehicle movements can be a potent tool for reducing the likelihood of collisions and enhancing overall safety.

Unlike reactive approaches that respond after an incident, proactive methods leverage sensor technologies, real-time data, and predictive analytics to anticipate and mitigate risks before they escalate. To develop such a proactive safety application, digital twin (DT) technologies offer a necessary framework. A DT is a real-time, synchronized virtual representation of assets and systems in the real world [4]. The flow of information can occur in both the directions, from physical to virtual and from virtual to physical, in a DT system. In the context of transportation, such a system is referred to as a transportation DT (TDT), where the physical components or processes of the transportation system have a real-time, synchronized, virtual representation. Among the many types of TDT, predictive DT (PDT) is a class of systems that go beyond representing the real-time state of a physical object; it uses complex modeling to simulate possible future states of the objects [5], [6]. In this study, we leverage a PDT framework to develop a proactive safety system tailored to complex, dynamic work

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zone scenarios. The goal is not merely to replicate the present state of traffic flow and vehicle behavior, but to anticipate potential future crashes before they materialize and enable timely interventions.

This study presents an infrastructure-enabled proactive work zone safety warning system that leverages a PDT framework, integrating real-time multisensor data, detailed high-definition (HD) maps, and a trajectory prediction model based on a historical prediction attention mechanism (AM). Specifically, the contributions of this study are in twofolds: 1) developing an infrastructure-enabled vehicle trajectory prediction framework that leverages the historical prediction network (HPNet) [7] model integrated with Lanelet2 HD maps and 2) developing a proactive safety warning generation application that incorporates probabilistic conflict modeling and vehicle bounding-box representations. HPNet is a deep neural network architecture designed to learn from multimodal trajectory data while accounting for spatiotemporal interactions among vehicles. By capturing both individual vehicle dynamics and intervehicle dependencies, HPNet generates future trajectory predictions with high temporal consistency and spatial precision, aligning well with the prediction capability requirements of a PDT system. Notably, to the best of our knowledge, this study is the first to adopt HPNet for infrastructure-based trajectory prediction, demonstrating that a self-driving car-based trajectory prediction algorithm can be effectively repurposed to operate from the vantage point of roadside infrastructure-mounted sensors. In addition, our proactive safety warning generation application validates the effectiveness of the PDT framework in issuing alerts for potential vehicle conflicts. For the development and evaluation of the PDT, we leverage a cosimulation approach that tightly integrates simulation of urban mobility (SUMO) with the CAR learning to act (CARLA) simulator. This cosimulation approach allows us to synthesize diverse, realistic vehicle trajectory datasets across varied traffic conditions, roadway configurations, and driver behaviors commonly observed in work zones.

II. Related Work

This section is divided into twofolds: 1) infrastructure-based vehicle trajectory prediction and 2) proactive safety warning generation and risk estimation.

A. Infrastructure-Based Vehicle Trajectory Prediction

Vehicle trajectory prediction is imperative for many safety-critical applications. The premise of the trajectory prediction problem is that given past motion states of a vehicle, estimate the future positions of that vehicle. These motion states typically include kinematic variables, such as position, velocity, and acceleration. The prediction algorithms may also use other relevant information, such as historical driving behavior or a detailed map of road geometry. Recent studies have highlighted that infrastructure-based trajectory prediction is tightly coupled with the quality of upstream vehicle tracking and trajectory reconstruction [8], [9]. For large-scale roadside deployments, raw trajectories are often fragmented by sensor noise, occlusions, and handovers between sensing regions. Recently, Liu et al. [8] proposed a multiaccess edge computing (MEC)-based framework that performs real-time trajectory calibration, cross-sensor fusion, and short-term trajectory prediction. Short-term prediction typically refers to 3–6 s of prediction horizon in the context of vehicles [10]. This framework demonstrates the scalability and feasibility of infrastructure-based tracking and short-term prediction.

There are several different approaches to predict vehicle trajectories, which can be broadly classified into data-driven approaches [11] using machine learning and deep learning, probabilistic models [12], Kalman filters and Bayesian estimators, and physics-based kinematic models [13]. In this study, we refer to the kinematic and probabilistic models as traditional approaches to trajectory prediction.

Traditional approaches typically overlook intervehicle interactions that influence driver behavior [14]. Another important factor guiding the vehicle's trajectory is the road geometry. The geometry of the road influences drivers as they negotiate roadway curves and turns. Considering the road geometry and how vehicles interact with the road shape and topology provides important cues for the prediction model. A few traditional approaches consider road geometry as boundary conditions; however, they lack semantic understanding of a roadway network [15].

Early data-driven methods used recurrent neural networks (RNNs), such as long short-term memories (LSTMs), to model trajectories [16], leveraging their strength in modeling temporal dependencies. Temporal convolutional networks (TCNs), an architecture that adapts convolutional neural networks (CNNs) for sequential data and integration with an AM [17], have shown improvements in trajectory prediction [18]. An AM in neural networks allows the model to focus on the most relevant parts of the input rather than treating all parts equally. The ability to focus on relevant parts of the input allows neural networks to achieve "contextual understanding," which has led to their use and integration with a wide range of deep-learning-based models [19].

In addition to sequential or temporal modeling, recent data-driven deep learning approaches often incorporate map information to improve understanding of road geometry and provide useful contextual cues. Two different approaches that include map information in trajectory prediction models are: 1) rasterized representations of roadway sections and 2) detailed vectorized maps that encode lane topology as a graph structure. Early efforts in this area used CNNs on rasterized roadway images, with vehicle trajectories encoded in the image. These methods demonstrated improved prediction accuracy compared with models that rely solely on sequential trajectory data [20]. More recent approaches leverage HD vector maps [21] that provide rich geometric and topological information, such as lane boundaries, connections, and traffic rules. These vector maps are typically encoded as graph structures, where lane segments are nodes, and their connections, including merging, branching, and priority relationships, are edges. Graph structure makes it compelling to apply graph neural networks (GNNs) and graph convolutional networks (GCNs) to trajectory prediction. The trajectory prediction model LaneGCN, introduced by Liang et al. [22],

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is built on this idea and has shown improved trajectory prediction, providing a foundation for subsequent studies. Studies such as VectorNet [23], QCNet [24], and HiVT [25] have demonstrated significant advances in trajectory prediction by leveraging HD vector maps, AMs, and GCNs. In addition, increased industry partnerships and benchmark datasets, such as Waymo [26], Argoverse [27], and INTERACTION [28], have accelerated the development of better-performing trajectory prediction models and mechanisms for comparing model results. Despite the benefits of incorporating HD map context in state-of-the-art trajectory prediction models, most existing infrastructure-based prediction studies continue to rely primarily on temporal modeling formulations. Representative approaches include recurrent or hybrid temporal models for roadside multiagent forecasting [29], [30], transfer-learning-based predictors adapted for roadside deployment [31], and transformer-based models that emphasize interaction modeling without explicit lane-level map encoding [32]. As a result, the rich map-centric representations that have driven recent advances in trajectory prediction remain comparatively underexplored in current infrastructure-based prediction pipelines.

Most trajectory prediction benchmark datasets primarily target self-driving vehicle applications, where a vehicle must anticipate the movements of surrounding vehicles to support its path-planning module and safety. This incentivizes the development of sophisticated trajectory prediction models, which are predominantly evaluated in self-driving contexts. It is also important to distinguish trajectory prediction from trajectory planning, as these two problems are often conflated in the literature. Trajectory planning methods aim to generate a feasible and optimal future path for an ego vehicle under dynamic conditions and safety constraints, assuming known intent and controllability. For example, the road obstacle-involved trajectory planner (ROITP) explicitly incorporates road obstacles, vehicle rollover dynamics, and stability constraints to generate safe trajectories for autonomous trucks using optimization-based formulations [33]. While such planning-oriented approaches are critical for autonomous driving, they differ fundamentally from trajectory prediction, which seeks to infer the future motion of surrounding vehicles whose intentions and control inputs are unobserved.

Recently, HPNet [7] ranked first on the INTERACTION dataset in the trajectory prediction task, due to its novel triple-factorized attention and the use of historical predictions to improve temporal consistency. However, the advances in prediction model architectures are also applicable to other domains, particularly roadside infrastructure-based trajectory prediction for safety applications. A key strength of HPNet's multimodal output is its ability to capture the inherent variability in human driving behavior. Human drivers exercise free will, often making subtle, discretionary choices that result in slightly different yet equally plausible trajectories. For instance, a driver may choose to navigate a work zone by slightly adjusting their path or speed, with each choice having a comparable likelihood. Chen et al. [34] point out the variability introduced by human behavior and different driving styles in decision-making environments. HPNet's multimodal predictions reflect this by generating multiple trajectory hypotheses, each adhering to physical constraints and map context, such as lane boundaries and temporary signage. These trajectories, while distinct, show high agreement due to the constrained nature of work zone environments, where variability is subtle and bounded by roadway geometry and traffic rules. This balance allows HPNet to model the nuanced decision-making of human drivers while maintaining temporal coherence and spatial consistency across predictions. The primary contribution of this study is to demonstrate the feasibility and effectiveness of using HPNet for infrastructure-based trajectory prediction in a roadway work zone setting.

B. Risk Estimation and Proactive Warning Generation

Accurate estimation of vehicle conflict risk and its translation into effective warnings remain central challenges in proactive transportation safety systems. To this end, recent research has increasingly adopted probabilistic modeling frameworks to address the uncertainties inherent in vehicle trajectories, driver behaviors, and dynamic traffic contexts such as highway work zones. A foundational principle in modern risk estimation is trajectory-level probabilistic modeling. Gaussian processes and hidden Markov models have been combined to predict vehicle motion under uncertain conditions, producing collision risk estimates that generalize beyond fixed assumptions of linearity or constant speed [35]. Similar probabilistic approaches have been applied in pedestrian-vehicle contexts, where Gaussian process regression and maneuver likelihoods allow for granular conflict quantification with lower computational costs than deep learning models [36]. These frameworks outperform traditional surrogate safety measures (SSMs), such as time-to-collision (TTC), which often rely on rigid motion assumptions.

Although postencroachment time (PET) and TTC remain fundamental to SSM approaches, many researchers have enhanced their predictive capability by integrating these indicators within probabilistic modeling frameworks that better capture traffic's inherent randomness. Among these, extreme value theory (EVT) has emerged as a powerful tool for quantifying rare but critical safety events [37], [38], [39], [40], [41], [42], [43]. Alozi and Hussein [44] analyzed large-scale AV-pedestrian interaction datasets from platforms, such as Motional and Lyft, using PET and TTC to derive risk profiles. Their findings indicated an elevated risk of collisions in the early stages of autonomous vehicle deployment, with an estimated 4–5.5 collisions per million vehicle-kilometers traveled (VKT). When contextual variables were added, this range narrowed to 2.3–3.7 per million VKT, offering greater statistical confidence. Their results underscore how EVT-based probabilistic models can accommodate behavioral variability and provide scalable risk metrics in rapidly evolving transportation systems. A recent framework integrates backward and stochastic forward reachable sets to distinguish guaranteed safe states from probabilistically risky ones, thereby enhancing both safety guarantees and precision in dynamic conflict identification [45]. Such hybrid techniques help mitigate overconservatism in earlier reachability analyses, while supporting confidence-aware risk predictions essential for real-world deployment. On the other hand, Jiao et al. [38]

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developed a probabilistic model that characterizes traffic conflicts as context-sensitive extremes in road-user behavior. Unlike conventional SSM approaches that use static thresholds based on conflict type, their method reframes conflict risk estimation as a statistical learning problem, in which extreme value inference enables more adaptive detection. To develop a proactive safety application, we leverage a probabilistic model that characterizes traffic conflicts and present a mathematical framework for predicting pairwise vehicle conflicts based on multistep trajectory forecasts.

III. PDT Framework

Fig. 1 presents a PDT framework for proactive work zone safety using roadside sensor data, which comprises three layers: 1) physical world; 2) PDT; and 3) communication gateway. The physical world serves as the bottom layer and encompasses all the transportation-related physical objects, such as vehicles, work zone road segments, workers, and roadside sensors. The top layer represents the PDT, which consists of digital shadow, digital sibling, and real-time proactive safety applications. The top and bottom layers are connected via the communication gateway, enabling real-time data exchange between them. Roadside infrastructure can be equipped with computing devices and sensors (e.g., cameras, radar, and LiDAR) to precisely detect vehicles and workers and continuously generate their trajectories within the work zone roadway segment. PDT layer consists of digital shadow, digital sibling, and proactive safety applications. In the context of PDT, a digital shadow is a digital replica of vehicles' and workers' trajectories. Digital shadows can be created and updated in real-time by aggregating, synchronizing, and fusing data from the physical world. Digital shadow receives sensor data from the physical world, processes it, and stores the processed data in the digital sibling database, but it cannot transmit information back to the physical world.

Digital sibling has four basic components: 1) database; 2) Lanelet2 HD Map; 3) HPNet trajectory prediction model; and 3) predicted trajectories (PTs) for all vehicles and maneuver scenarios. The prediction model can predict all plausible vehicle trajectories, each with a different probability. Finally, the proactive safety application module calculates conflict probability for all combinations of PT pairs and uses a conflict decision threshold to generate proactive safety alerts. Sections III-A–III-C detail the development pipeline, prediction model, and proactive safety application.

A. Integration of Simulation, Prediction, and Warning Generation

This study presents a comprehensive pipeline for developing a proactive warning system for work zones within a PDT framework, integrating scenario design, cosimulation, model training, and warning generation, as illustrated in Fig. 2. The process begins with the design stage, where road networks and traffic scenarios are created using tools, such as RoadRunner [46] and Commonroad Scenario Designer [47]. These generate OpenDrive and Lanelet2 HD maps, along with SUMO network and route files, and support microscopic traffic flow models tailored for realistic simulations. At the cosimulation stage, the SUMO and CARLA simulators interact to exchange vehicle and traffic-light states, creating a realistic environment for data collection and testing that mirrors real-world implementation processes. The prediction model training stage involves generating trajectory datasets from simulations, preprocessing them, and training a trajectory prediction model. Finally, the proactive warning generation stage uses the trained model for online inference, enabling real-time deployment by predicting future trajectories and generating proactive warnings using vehicle data.

B. Trajectory Prediction of All Plausible Future Maneuverings

In this study, we used HPNet, developed by Tang et al. [7]. The key contribution of the authors is the development of temporally consistent trajectory prediction of all plausible future maneuvers. Many conventional trajectory prediction models treat each future timestep independently, leading to temporally incoherent predictions where the sequence of predicted positions lacks smoothness or dynamic feasibility. Such framewise decoupling results in discontinuities in velocity and heading, particularly under multimodal settings where multiple hypotheses are generated. HPNet mitigates this issue through a hierarchical decoder that models future trajectories as temporally dependent sequences. Let the observed trajectory of a vehicle be denoted as $\mathbf{X}_{1:T_{\text{obs}}} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_{T_{\text{obs}}}\}$, where T_{obs} is the number of historical timesteps, and $\mathbf{x}_t \in \mathbb{R}^2$ represents the 2-D position at time t. The goal is to predict K plausible future trajectories $\{\hat{\mathbf{Y}}^{(k)}\}_{k=1}^{K}$, where each $\hat{\mathbf{Y}}^{(k)} = \{\hat{\mathbf{y}}_1^{(k)}, \hat{\mathbf{y}}_2^{(k)}, \ldots, \hat{\mathbf{y}}_{T_{\text{pred}}}^{(k)}\}$ is a temporally ordered sequence of future positions. Specifically, the model autoregressively decodes future positions, where the prediction at timestep $t+1$ is conditioned not only on the historical context $\mathbf{X}_{1:T_{\text{obs}}}$ but also on the predictions up to timestep t, i.e., $\hat{\mathbf{Y}}_{1:t}$. This design enforces internal temporal consistency and ensures that trajectory evolution follows physically plausible motion patterns. Moreover, HPNet's multihead attention modules, such as historical and mode attention, allow the model to selectively focus on salient temporal cues and disambiguate agent intent across the full prediction horizon.

The HPNet model was originally developed for an egocentric self-driving vehicle that needs to assess the future trajectories of surrounding vehicles. Unlike its initial application with onboard sensors, our implementation adopted HPNet architecture for roadside infrastructure equipped with stationary LiDAR, radar, and/or camera sensors in a work zone scenario. HPNet uses a hierarchical, spatial–temporal deep learning framework to extract and model complex interactions and movement patterns. At its core, the network integrates GNNs, specifically using graph attention networks (GATs), to encode spatial and temporal interactions between vehicles and the roadway section. It processes the HD Map as a vector graph with attention modules. This graph-based representation enables HPNet to dynamically emphasize relevant spatial relationships and temporal dependencies, thereby improving the model's predictive performance. Multiple attention layers dynamically weight spatiotemporal features, which enables

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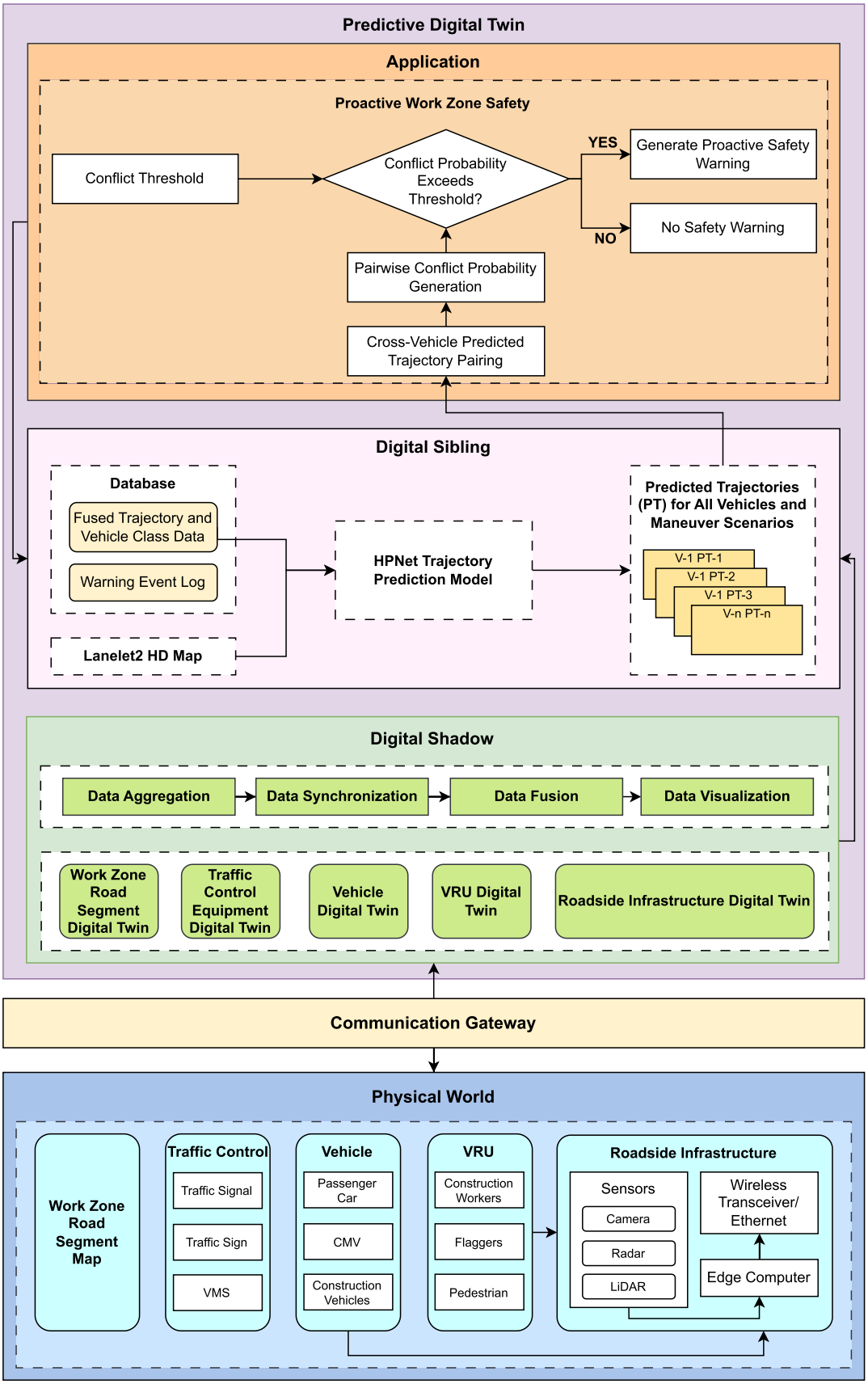


Fig. 1. PDT framework for proactive work zone safety.

multimodal trajectory predictions that account for inherent uncertainties in future vehicle movements.

AM [17] is a core component of the model. AMs enable the model to selectively focus on relevant parts of the input

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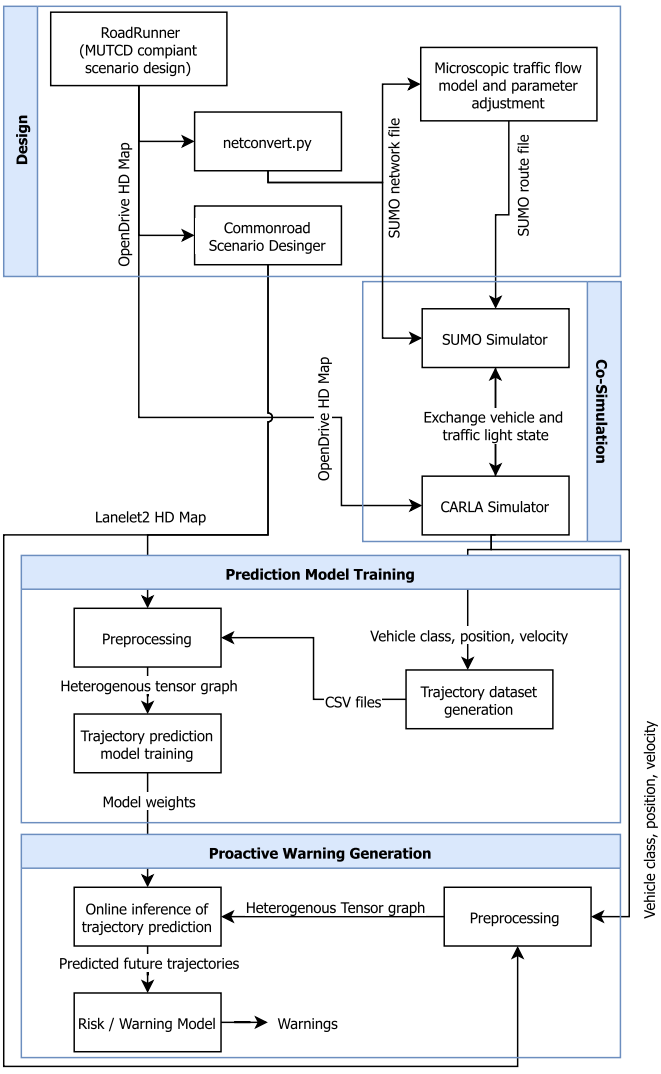


Fig. 2. Framework integrating simulation, prediction, and warning generation under a PDT framework.

data when generating predictions, significantly enhancing predictive accuracy and contextual awareness. The AM operates on three fundamental parameters: query (Q), key (K), and value (V). Their roles are as follows: query (Q) represents the elements seeking information. Key (K) represents the actual information content provided in response to the queries, weighted by relevance, determined by query–key dot-product similarity. The input information is tokenized by the embedding process and used as the K, Q, and V matrices. The model inputs include vehicle trajectories (sequences of position, heading, and velocity), which are encoded by a multilayer perceptron (MLP) into location-dependent vehicle feature embeddings (`E_a`). In addition, static attributes, such as vehicle length, width, and type, are incorporated to enrich the representation. Map attributes, including lane boundaries and center lines, are encoded into map feature embeddings (`E_m`). Relative features, capturing spatial relationships, such as distances, angles, and heading, and differences between agents and map elements, are encoded into relative feature embeddings (`E_e`).

The model architecture, as depicted in Fig. 3, processes the inputs through a series of attention-based layers to generate and refine multimodal trajectory predictions. The process begins with learnable mode tokens, representing K possible future trajectories for each agent, which act as queries in AMs. Temporal attention uses multihead cross-attention, with mode tokens as queries and vehicle/relative embeddings (`E_a`, `E_e`) as keys and values, to capture historical vehicle dynamics over time. Spatial attention uses cross-attention with mode tokens as queries and map/relative embeddings (`E_m`, `E_e`) to integrate roadway context, aligning predictions with lane constraints. Mode attention further refines predictions through three sublayers: agent attention (`P_a`) models interagent interactions using self-attention on prediction embeddings (`P`) and relative features; historic prediction attention (`P_hp`) ensures temporal coherence by capturing dependencies within PT; and mode attention disambiguates multiple plausible trajectory hypotheses by modeling interactions across modes.

The trajectory prediction process unfolds in two stages: proposal generation and refinement. In the proposal stage, an MLP (`traj_propose`) decodes mode embeddings (`m_embs`) into initial multimodal trajectory proposals (`traj_propose`) in global coordinates, with shape $[(N_1, \ldots, N_b), H, K, F, 2]$, where $N_1, \ldots, N_b$ denote agents per batch, H is the historical timestep, K is the mode, F is the future timestep, and 2 represents 2-D coordinates. These proposals define anchor points by selecting the middle future timestep ($F//2$) to compute anchor positions (`n_position`) and headings (`n_heading`), which are transformed into local coordinates and encoded into anchor embeddings (`n_embs`) using an MLP (`proposal_to_anchor`). In the refinement stage, anchor embeddings undergo another temporal attention stage to integrate historical agent data, a spatial attention stage to incorporate map context, and a mode attention stage to refine interactions across agents, historical steps, and modes. An MLP (`traj_refine`) generates a correction term (`traj_refine`), which is added to the initial trajectory proposal and transformed back to global coordinates using `transform_traj_to_global_coordinate`, yielding the final output (`traj_output`). This two-stage approach ensures contextually informed and temporally consistent predictions, overcoming the limitations of framewise decoupled models.

C. Proactive Safety Warning Generation

In this section, we present a mathematical framework for predicting pairwise vehicle conflicts based on multistep trajectory forecasts. Each vehicle is represented geometrically as a rigid rectangular bounding box situated in a 2-D Euclidean space. The center of the vehicle is denoted by a time-indexed sequence of predicted positions $(x_i^{(k)}, y_i^{(k)})$, accompanied by an orientation angle $\psi_i^{(k)}$, where k indicates the prediction step and i indexes the vehicle within the set of vehicles present in a given time frame. The shape of each vehicle is defined by parameters for length, l, and width, w, and the corresponding bounding box is constructed by applying a planar rotation matrix to a predefined set of local coordinates to represent the

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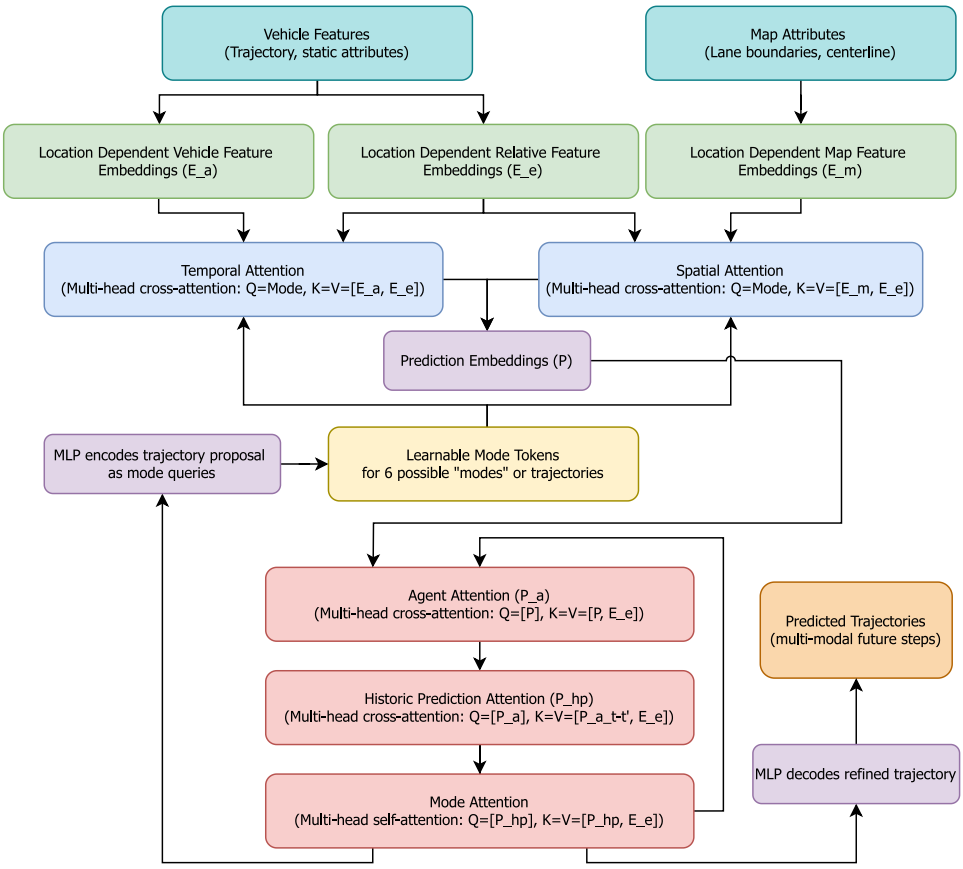


Fig. 3. Historical trajectory prediction model architecture with multistage attention for temporally consistent trajectory prediction.

corners and midpoints of the vehicle's physical envelope. The transformation into the global coordinate system is performed by rotating these points using $\psi_i^{(k)}$ and translating them to the global position $(x_i^{(k)}, y_i^{(k)})$. This transformation for a point (u, v) in the vehicle's local coordinate frame is expressed as $(x_i^{(k)}, y_i^{(k)}) + \mathcal{R}(\psi_i^{(k)}) \cdot (u, v)^T$, where $\mathcal{R}(\psi)$ denotes the standard 2-D rotation matrix. The notation $(u, v)^T$ denotes the column vector form of a point defined in the vehicle's local coordinate frame. While (u, v) represents a 2-D point as a row vector, its transpose $(u, v)^T = \begin{bmatrix} u \\ v \end{bmatrix}$ is required for matrix multiplication with the 2×2 rotation matrix $\mathcal{R}(\psi)$. This transformation rotates the local point according to the vehicle's orientation ψ to be accurately mapped into global coordinates

$$\mathcal{R}\left(\psi_i^{(k)}\right)\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} \cos\psi & -\sin\psi \\ \sin\psi & \cos\psi \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} u\cos\psi - v\sin\psi \\ u\sin\psi + v\cos\psi \end{bmatrix}. \quad (1)$$

For each discrete time frame and prediction step k, we evaluate all the unique unordered pairs of vehicles (v_i, v_j), with $i < j$, to estimate the likelihood of a conflict. The minimum Euclidean distance between any pair of transformed bounding-box points associated with v_i and v_j is computed and denoted as $d_{ij}^{(k)}$

$$d_{ij}^{(k)} = \min_{a,b} \left\| p_{ia}^{(k)} - p_{jb}^{(k)} \right\|_2 \quad (2)$$

where $p_{ia}^{(k)}$ and $p_{jb}^{(k)}$ are the transformed global coordinates of the ath and bth bounding-box points for vehicles i and j, respectively. In a conflict hierarchy, the closer the proximity between approaching road users (e.g., vehicles), the less safe they are and the higher the probability of a potential collision [38]. Hence, the scalar distance $d_{ij}^{(k)}$ is subsequently transformed into a proximity-to-risk map via an exponential decay function of the form

$$P_{ij}^{(k)} = \exp\left(-\frac{d_{ij}^{(k)}}{\lambda}\right) \quad (3)$$

where $\lambda > 0$ is a decay constant governing the rate at which conflict likelihood diminishes with spatial separation. This function handles the conflict probability approaching unity as the separation approaches zero and decaying asymptotically toward zero as the distance increases, while providing a smooth, differentiable risk measure based on physical proximity. Pairwise interaction of vehicles is monitored for potential conflicts if the distance $d_{ij}^{(k)}$ falls below a distance threshold δ_d. As conflict risk is continuously calculated, a proactive warning is generated if risk estimation exceeds a confidence threshold $\tau \in (0, 1)$. The exponential model provides a decision surface where τ and λ parameters jointly determine a clearance contour, which delineates the boundary between safe and potentially conflicting vehicle interactions. The τ is treated as a warning-generation sensitivity parameter, λ controls the decay scale of the exponential model, and δ_d

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serves as a reference clearance for labeling a conflict case. These parameters provide a mechanism for calibrating decision policy and enable sensitivity or stability analysis. SUMO's built-in SSM device logs time- and position-stamped "close-call" events based on standard SSM indicators (e.g., TTC, hard-braking event) during simulation runtime, which provide a principled starting point defining reference clearance level δ_d and for calibrating the $(\delta_d, \lambda, \tau)$ parameter set.

The computational process iterates over all time frames and prediction horizons, maintaining a structured record for each evaluated vehicle pair: their geometric separation, the associated conflict probability, and whether the pair meets the criteria for conflict or high-risk designation. It has directional sensitivity in conflict modeling. For example, vehicles approaching each other head-on or at acute angles may present higher conflict risk despite moderate centroidal distances. The alignment of bounding-box surfaces implicitly captures such configurations for a more accurate detection of spatial convergence zones. This is especially critical in environments with complex maneuvers, such as work zones, intersections, or lane merges, where lateral and angular interactions are dominant. From a mathematical standpoint, bounding-box distances generalize center-based metrics by reducing to the latter in the degenerate case where vehicle dimensions reduce to zero. As such, they preserve the analytical simplicity of Euclidean norms while enriching the geometric expressiveness of the model.

IV. SUMO-CARLA Experimental Setup

In this study, we use the SUMO and CARLA simulators, widely accepted open-source tools for traffic, vehicle dynamics, and sensor simulation. SUMO provides realistic large-scale traffic simulation capabilities using standard microscopic traffic flow models and is particularly adept at simulating detailed merging behavior in constrained environments, such as work zones [48]. In this study, CARLA complements SUMO by providing high-fidelity sensor simulation, including lidar, radar, and camera sensors. We adjusted the car-following and lane-changing models in SUMO to closely match observed real-world driver behavior within work zones. These behaviors include realistic acceleration, deceleration, and merging maneuvers validated by traffic flow studies [49].

The merging behavior approaching a lane closure is particularly important in work zone scenarios. By default, SUMO uses the LC2013 [50] lane-changing model to make a decision to change lane at every simulation step. This model considers four different motivations to make lane changes, namely: 1) strategic change; 2) cooperative change; 3) tactical change; and 4) regulatory change. The lane change due to the lane closure ahead is a strategic change. The default behavior of this lane-changing model in SUMO is that lane change happens in an instant. When the model satisfies all the criteria for changing lanes, the vehicle moves to the target lane in an instant. Instant lane changes are prohibitive as it is necessary to capture the lane-changing dynamics for our application. To better suit our study, we use SUMO's sublane model, an extension of the LC2013 lane-changing logic that allows continuous lateral movement across the width of a lane rather than abrupt, instantaneous jumps. This sublane model, referred to as SL2015, provides a more realistic representation of lane-changing behavior by allowing vehicles to gradually shift their lateral positions over time. In the sublane model, a road is conceptually divided into many fine-grained lateral subdivisions (sublanes), allowing vehicles to occupy fractional positions within a lane or between lanes. This enables modeling lane changes as a time-dependent process influenced by vehicle dynamics rather than as a discrete event. To activate the sublane model, SUMO requires to be configured with the `-lateral-resolution <FLOAT>` parameter set to a value smaller than the lane width, which is 0.25 m in our case. With the sublane model enabled, SUMO switches the lane-changing model from the default LC2013 to the SL2015 model, which can use lane subdivisions for smoother, more realistic transitions. Once enabled, the vehicles can initiate a lane change by steering toward the target lane, and the movement unfolds gradually over multiple simulation steps. In addition to LC2013 model parameters, the trajectory and duration of this lateral transition are governed by additional vehicle parameters, such as lcSublane, lcAssertive, lcPushy, lcImpatience, lcTimeToImpatience, lcAccelLat, latAlignment, and lcMaxSpeedLatStanding. These parameters together shape how vehicles behave during lane changes. For instance, lcSublane controls the vehicle's preference for staying aligned laterally within a lane, while lcAssertive and lcPushy determine how willing a vehicle is to accept tighter gaps or initiate lane changes in partially blocked spaces. lcImpatience dynamically scales the assertive and pushy tendencies as time progresses, and lcTimeToImpatience measures the time it takes to reach peak impatience. lcAccelLat restricts the lateral acceleration to reflect realistic dynamics, and latAlignment allows vehicles to aim for specific offsets within the lane (e.g., center, left, or right). Finally, the parameters lcMaxSpeedLatStanding and lcMaxSpeedLatFactor control the lateral speed envelope, scaling lateral velocity based on vehicle speed to prevent unrealistic side movements. Together, these parameters enable nuanced, gradual, and context-aware lateral maneuvers that are critical for simulating realistic merging behavior in work zones where vehicles should respond to upstream lane closures. This modeling enhancement is especially useful because it allows us to accurately capture the interactions between merging vehicles and through traffic, as well as model behaviors such as hesitation, courtesy yielding, and multivehicle negotiations near the taper zone of the work zone. In addition, the continuous, dynamics, and time-evolving lateral motion representation allows for better integration with trajectory prediction algorithms, which requires smooth and physically plausible vehicle motions.

The road network for SUMO-CARLA cosimulation is created using MathWorks Roadrunner software, following manual on uniform traffic control devices (MUTCDs) guidelines for lane closure in a work zone, including speed reduction and taper lengths [51]. Fig. 4 shows a portion of the work zone map created in Roadrunner. The map geometry can be exported in the OpenDrive (map.xodr) HD Map format, which is a direct input to the CARLA simulator. CARLA supports native

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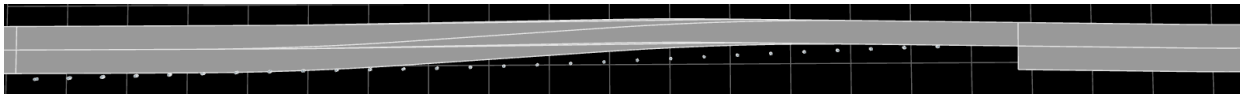


Fig. 4. Work zone scenario with lane closure and taper designed in RoadRunner.

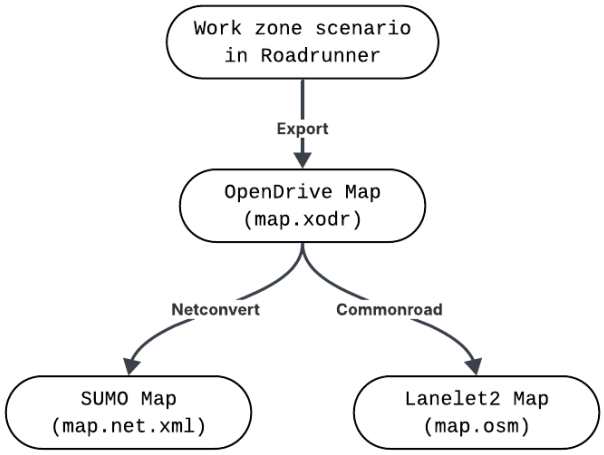


Fig. 5. Map conversion workflow from OpenDRIVE to SUMO and Lanelet2 formats.

parsing of OpenDrive-format road networks. CARLA uses the file and constructs corresponding 3-D meshes for the road topology and renders in 3-D simulation environment. However, the SUMO simulator cannot take an OpenDrive map as input. It uses its own XML format for network definitions. On the other hand, the trajectory prediction module uses another HD Map format, Lanelet2. Fig. 5 illustrates how the map has been converted to be used by SUMO, CARLA, and the trajectory prediction module.

To facilitate SUMO-CARLA cosimulation, SUMO's `netconvert` tool was used for converting OpenDrive maps into SUMO-compatible networks. `Netconvert` parses the road geometry and attributes from the XODR format and generates an equivalent network topology interpretable by SUMO. The "net.xml" file serves as the road network for SUMO, enabling us to simulate traffic and synchronize it with CARLA through our cosimulation bridge. This allows us to simulate traffic interactions, including vehicle behavior through the work zone. Realistic traffic modeling is performed by SUMO, while CARLA provides a realistic 3D environment so that the environment can be perceived by sensors such as LiDAR, radar, and camera.

The adopted HPNet model has a module that parses the Lanelet2 format HD Map to create its internal vector graph representation of the map. We converted the OpenDrive format map into Lanelet2 using the open-source tool Commonroad Scenario Designer [47]. Commonroad provides a common abstraction for elements commonly found in different HD Map standards, which in turn enables seamless conversion between standard map formats.

V. Data Generation

Vehicle trajectories are derived from infrastructure-mounted LiDAR, radar, and camera sensors using off-the-shelf detection and tracking components. Camera-based vehicle detection is performed using a YOLO-11 object detector, LiDAR point clouds are processed using standard Euclidean-clustering-based detection, and radar provides complementary range and radial velocity measurements. Camera detections are projected onto the roadway using a planar homography under a locally planar road assumption, while LiDAR and radar detections are transformed into a common stationary world coordinate frame via rigid extrinsic calibration. All the world-frame coordinates are expressed in a projected metric system using the universal transverse mercator (UTM) projection to ensure global-coordinate consistency across sensors, trajectories, and HD maps. Following spatial alignment, detections are temporally associated using a SORT-based multiobject tracking algorithm, and trajectory-level fusion is performed using an extended Kalman filter (EKF) to jointly estimate vehicle position, velocity, and heading over time. The resulting fused trajectories are preprocessed to match HPNet's input format, including fixed-length historical sequences, agent attributes, and alignment with the Lanelet2 HD map, enabling infrastructure-based trajectory prediction. All three sensor modalities collect data at a frequency of 10 Hz, which facilitates high temporal resolution and continuous vehicle tracking. In our simulation setup, we also collect ground-truth data for all vehicles at 10 Hz (0.1-s time interval), which we use to train the HPNet model.

The structure of the INTERACTION dataset [28] served as a reference when building our dataset from simulation data. The dataset consists of comma-separated-value (CSV) files, each containing a 4-s-long scenario. Since the data are recorded at 10 Hz, and there are 18–22 vehicles at any given time in the simulation, each CSV file has approximately 800 rows of data. Data snapshots recorded at 10 Hz (100-ms intervals) are referred to as a frame. So, a 4-s scenario has 40 frames, where the first ten frames are considered as input and the subsequent 30 frames are ground-truth labels. A total of 20 000 s of driving simulation data were generated, which results in 5000 individual cases, each stored as a separate CSV file. These cases collectively represent approximately 5.56 h of continuous simulated driving. The CSV files are indexed by "case_id" (e.g., trajectory_data_case_10.csv). The description of the data fields is provided in Table I.

VI. Evaluation Outcomes

A. Evaluation Metrics

To evaluate the accuracy of PTs, we use two standard metrics widely used in the literature: average displacement error (ADE) and final displacement error (FDE). Let $\mathbf{y}_t = (x_t, y_t)$ denote the ground-truth position at time step t, and let $\hat{\mathbf{y}}_t = (\hat{x}_t, \hat{y}_t)$ represent the predicted position at the same time

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TABLE I

DESCRIPTION OF COLUMNS IN THE TRAJECTORY_DATA_CASE_XXX.CSV FILE

Column	Attribute	Description
1	track_id	The ID associated with each vehicle in the scene. The track_id ranges from 1 to n, where n is the number of vehicles in the scene.
2	timestamp_ms	The timestamp associated with the trajectory, ranging from 100 ms to at most 4000 ms, conforming to the 4-second trajectory recording.
3	frame_id	For each vehicle (track_id), the number of frames the vehicle appeared in the data. It ranges from 1 to at most 40, as a vehicle appears at most 40 times within the 4-second period recorded at 100 ms intervals.
4	agent_type	The category of the vehicle (e.g., car, truck).
5	x	The x position of the vehicle (meters) in the local coordinate system.
6	y	The y position of the vehicle (meters) in the local coordinate system.
7	vx	The x component of the velocity (meters per second) in the local coordinate system.
8	vy	The y component of the velocity (meters per second) in the local coordinate system.
9	psi_rad	The yaw angle heading of the vehicle (radians).
10	length	The length of the vehicle in meters.
11	width	The width of the vehicle in meters.

step. Given a prediction horizon of T time steps, the metrics are defined as follows:

$$\text{ADE} = \frac{1}{T}\sum_{t=1}^{T}\left\|\hat{\mathbf{y}}_t - \mathbf{y}_t\right\|_2 \tag{4}$$

$$\text{FDE} = \left\|\hat{\mathbf{y}}_T - \mathbf{y}_T\right\|_2. \tag{5}$$

The ADE measures the average Euclidean distance (L2 norm) between the predicted and ground-truth positions over the entire prediction horizon, providing a holistic view of the model's performance over time. In contrast, the FDE focuses solely on the Euclidean distance at the final prediction time step, capturing the model's accuracy in estimating the vehicle's ultimate destination. These metrics jointly capture both spatial consistency and endpoint-oriented accuracy. A low value of ADE and FDE is desired for a trajectory prediction model.

B. Training and Testing of Trajectory Prediction Model

We trained the HPNet model for 60 epochs using the Adam optimizer with an initial learning rate of $5e^{-4}$, decayed by a factor of $1e^{-4}$ after four warmup epochs. As shown in Fig. 6, the training loss drops sharply during the first ten epochs, from over 37 down to approximately 0.39, before decaying more gradually to a final value of 0.06. The validation loss follows a very similar trajectory, decreasing from an initial 31 to 0.08 by epoch 60. The close alignment of the two curves, especially beyond epoch 15, indicates that the model converges reliably without significant overfitting. The vertical axis of Fig. 6 is in logarithmic scale to capture the rapid initial convergence and the later fine-scale improvements.

We further evaluate predictive accuracy using two standard trajectory-prediction metrics: ADE and FDE. Fig. 6 also illustrates that the minJointADE decreases from roughly 14.0 m at epoch 1 to 0.13 m at epoch 60, while the minJointFDE falls from about 32 m down to 0.322 m over the same training horizon. Both the metrics stabilize after epoch 30, confirming that the model's improvements plateau once the network has sufficiently learned the underlying motion patterns in the dataset.

Overall, the parallel decline in training and validation losses, combined with the monotonic reduction in ADE and FDE,

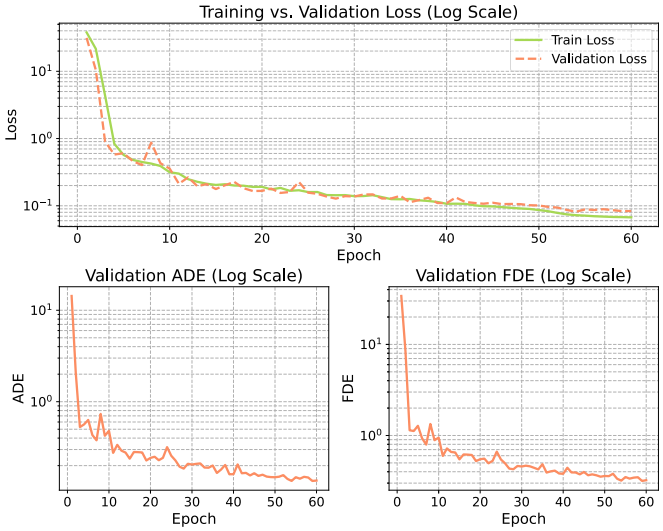


Fig. 6. Training and validation loss with ADE/FDE trends across epochs.

TABLE II

HPNET PERFORMANCE ON DIFFERENT DATASETS

Dataset	minJointFDE (m)	minJointADE (m)
Argoverse	1.0986	0.7612
Interaction	0.8231	0.2548
CARLA Town Simulation	0.5240	0.2354
Work zone Simulation	0.3228	0.1327

demonstrates that HPNet effectively captures vehicle motion characteristics. The absence of a widening gap between training and validation losses, along with the smooth decay of displacement errors, suggests generalization and supports the choice of training schedule and associated hyperparameters.

JointADE is the measure of the average L2-norm distance across all vehicles' predicted and actual trajectories, while minimum joint ADE (minJointADE) indicates the closest mode (of 6) to the ground truth. On the other hand, minimum joint ADE (minJointADE) measures the average L2-norm distance at the final time step for all vehicles. These metrics are widely used to benchmark trajectory prediction performance. Table II shows comparison of these metrics from our infrastructure-based work zone simulation dataset with

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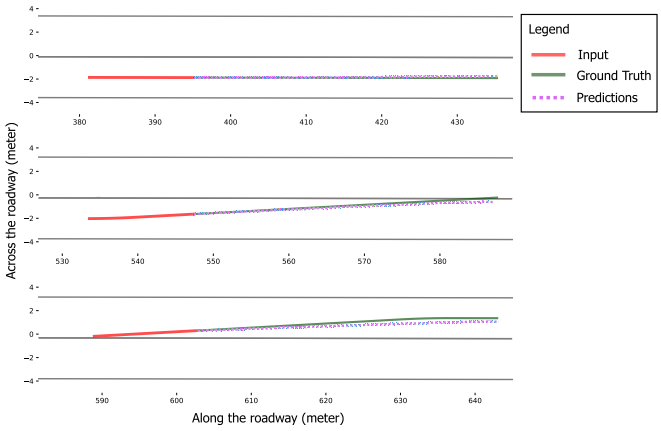


Fig. 7. Predicted versus ground-truth trajectories for individual vehicles.

the Argoverse and Interaction datasets. The HPNet model demonstrates superior performance on the work zone simulation dataset, achieving a minJointFDE of 0.3228 m and a minJointADE of 0.1327 m, significantly lower than the benchmarks on the Argoverse (minJointFDE: 1.0986 m, minJointADE: 0.7612 m) and Interaction (minJointFDE: 0.8231 m, minJointADE: 0.2548 m) datasets. This performance disparity can be attributed to the differences in dataset complexity. The benchmark datasets, Argoverse and Interaction, are derived from real-world data and capture the diverse and unpredictable behavior of human drivers. In contrast, the work zone simulation dataset, while instrumental in developing and testing the entire pipeline and designed to be as realistic as possible, relies on an underlying microscopic traffic flow model. This model inherently lacks the full spectrum of variability observed in real-world human drivers, which may lead to more predictable trajectories and improved model performance. On the other hand, the improved results may stem from the advantages of our roadside infrastructure-based setup, which offers a more favorable viewpoint with reduced occlusion compared with a self-driving car. This enhanced visibility likely contributes to the model's lower ADE and FDE, highlighting the potential of infrastructure-based systems for trajectory prediction.

In addition, the HPNet model was trained and evaluated on CARLA Town 06, which features a mid-density urban environment with multiple intersections and more diverse interaction scenarios than the work zone setup. As reported in Table II, the model achieves comparable performance on the CARLA town simulation dataset, indicating that the learned representations are not overly specialized to a single scenario. This consistency across environments suggests a reasonable level of generalization and supports the applicability of the proposed infrastructure-based prediction framework beyond a single work-zone configuration.

Fig. 7 illustrates the PTs for selected individual vehicles over a 4-s horizon, plotted alongside ground-truth tracks. Fig. 7 also covers both the in-lane driving and lane-changing cases. The predictions align closely with actual vehicle paths, capturing both the temporal evolution and the spatial curvature of motion. The multimodal nature of the prediction is evident in the closely clustered trajectories, which reflect subtle variations in potential driving behavior under the constraints imposed by work zone geometry.

TABLE III

CONFUSION MATRIX FOR CONFLICT WARNING GENERATION UNDER BASELINE CONFIGURATION $(\delta_d, \lambda, \tau) = (7, 7, 0.7)$

	Actual		
Predicted	Conflict	No Conflict	**Total**
Warning Issued	12 (TP)	24 (FP)	36
No Warning	42 (FN)	1,722 (TN)	1,764
Total	54	1,746	1,800

TABLE IV

PERFORMANCE METRICS DERIVED FROM CONFUSION MATRIX

Metric	Formula	Value
Conflict Prevalence	$\pi = N_+/N$	0.030 (3.0%)
Precision	$Pr = TP/(TP + FP)$	0.333
Recall (Sensitivity)	$Re = TP/(TP + FN)$	0.222
False Negative Rate	$FNR = FN/(FN + TP)$	0.778
False Positive Rate	$FPR = FP/(FP + TN)$	0.014
Specificity	$1 - FPR$	0.986

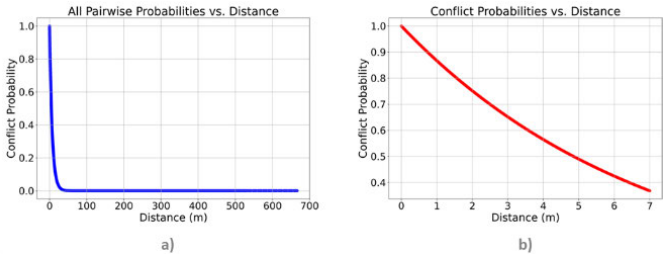


Fig. 8. Conflict probability trends between two selected target vehicles across future prediction steps approaching a work zone area. (a) Full distribution of pairwise conflict probabilities at all predicted time horizons, illustrating how risk evolves over distance. (b) Filtered view showing only instances where the minimum bounding-box distance falls below 7 m, highlighting high-risk interactions within safety-critical proximity.

C. Evaluation of Proactive Safety Warning Generation

The conflict risk estimation framework discussed earlier in Section III-C is applied to predicted vehicle trajectories generated over successive time horizons. A critical component of this application involved selecting appropriate decision parameter values to classify interactions as potential conflicts and to issue proactive driver warnings. The performance of this warning-generation module is evaluated by aggregating decisions across all time steps for 60 test cases comprising a total of 1800 predicted data frames. Under baseline operating configuration $(\delta_d, \lambda, \tau) = (7, 7, 0.7)$, determined through empirical tuning and refined by sensitivity analysis, the confusion matrix and the derived performance metrics are presented in Tables III and IV, respectively.

The conflict prevalence, defined as the proportion of frames containing a conflict event in the dataset, is $\pi = 0.03$, indicating that conflicts occur in approximately 3% of frames. This class imbalance reflects the inherently rare nature of safety-critical interactions in realistic traffic streams. Due to this imbalance, a hypothetical system that never issues warnings would achieve 97% accuracy simply by exploiting the

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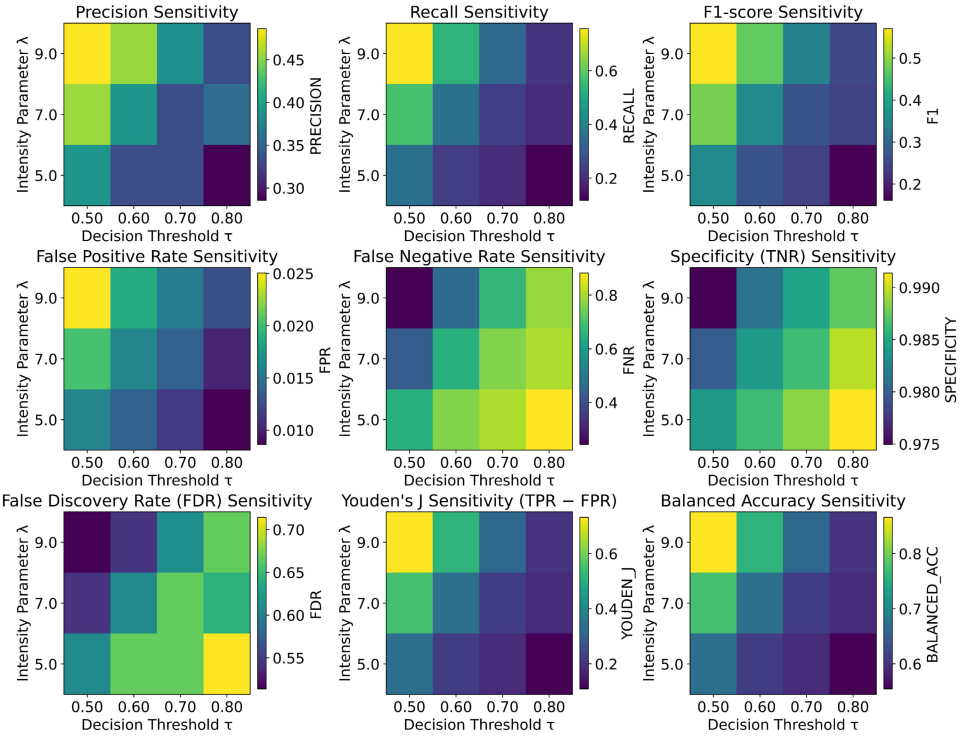


Fig. 9. Joint sensitivity of detection performance across the (λ, τ) parameter space.

class imbalance, while completely failing its safety function. Accuracy is, therefore, a misleading metric for this problem. Performance should instead be assessed using metrics such as precision, recall, and false-positive rate (FPR), which separately assess how the system handles conflict and nonconflict cases between vehicles.

At the $\tau = 0.7$ threshold, the system achieves a precision of 0.333, meaning roughly one-third of issued warnings prove correct. This reflects a deliberate design tradeoff. The system prioritizes avoiding false alarms over catching every potential conflict, as shown by the low FPR of 0.014. When no conflict is developing, the system incorrectly warns only 1.4% of the time. This restraint is critical for acceptance among drivers, since excessive warnings lead to alert fatigue and driver disregard of the system. The conservative strategy comes with a tradeoff: a recall (sensitivity) of 0.222 and a false-negative rate of 0.778. A lower threshold would raise more potential conflict warnings but would also generate many more false alarms. Subsequent analysis explores this tradeoff by varying values of the threshold τ.

Fig. 8 illustrates the relationship between the distance between vehicle pairs and the corresponding conflict probability across the future prediction window. Fig. 8(a) highlights how risk increases as a vehicle approaches another, while the filtered view in Fig. 8(b) emphasizes the interactions occurring within the reference clearance distance defined by δ_d. This visualization demonstrates that high-risk interactions are naturally concentrated in close-proximity regimes, validating the use of the clearance threshold δ_d as a reference scale rather than a collision indicator.

Fig. 9 summarizes the joint sensitivity of detection performance across the (λ, τ) parameter space under the reference clearance, δ_d. Each panel visualizes how key metrics evolve as the decay scale λ and the decision threshold τ are varied. Core detection quality metrics (precision, recall, and $F1$-score) are presented alongside error-related measures (i.e., FNR, FPR, and specificity) and derived decision metrics (i.e., false discovery rate (FDR), Youden's J statistic, and balanced accuracy) together revealing the effect of τ and λ on conflict detection. The FDR represents the proportion of warnings that turn out to be false alarms, providing an alternative view of precision (FDR = 1 – Precision). Youden's J statistic combines sensitivity and specificity (J = Sensitivity + Specificity −1), providing a single measure that balances detection of conflicts and avoidance of false alarms, with values ranging from 0 (no better than random) to 1 (perfect classification). Balanced accuracy averages performance across conflict and nonconflict classes, ensuring the evaluation is not dominated by the majority class. Color intensity denotes metric magnitude, highlighting regions of stability, degradation, and tradeoff across the parameter space. Fig. 9 shows that performance varies smoothly across parameter ranges ($\lambda = 5, 7, 9; \tau = 0.5, 0.6, 0.7, 0.8$), with the dominant sensitivity arising from τ, while changes in λ primarily adjust the sharpness of the proximity-to-risk mapping exponential function without introducing abrupt degradation or instability. A higher λ value causes the risk to decay more slowly with distance, making the system more sensitive to very close interactions while being less reactive to moderate separations. Importantly, broad regions of stable performance are observed, indicating that the decision surface is not brittle with respect to physical or probabilistic parameterization. The baseline configuration ($\lambda = 7, \tau = 0.7$) sits in a stable region where small parameter perturbations do not cause dramatic performance shifts.

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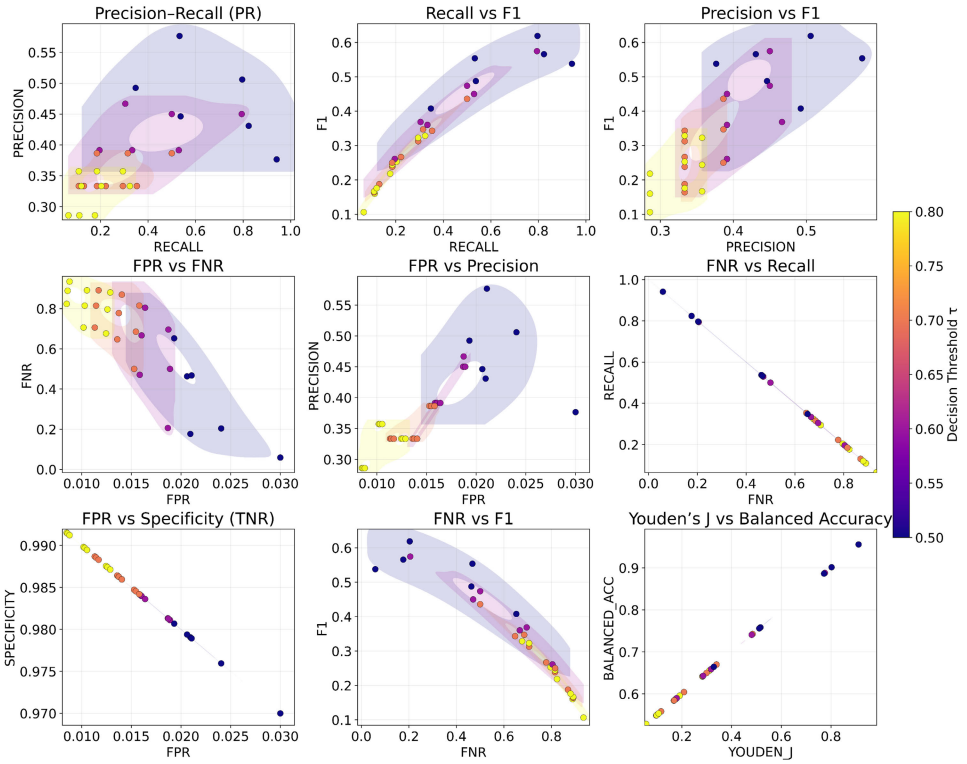


Fig. 10. Threshold-conditioned concentration patterns across multiple performance tradeoff spaces.

Fig. 10 provides a complementary view by illustrating threshold-conditioned concentration patterns across multiple performance tradeoff spaces. Each panel is a scatter plot, with each point representing a tested parameter configuration. Points are colored by threshold value τ, ranging from yellow ($\tau = 0.80$, conservative) to blue ($\tau = 0.50$, aggressive). The shaded regions show where parameter configurations create clusters. The precision–recall plot shows the fundamental tradeoff. Precision is the fraction of correct warnings, while recall is the fraction of conflicts detected. The FPR versus FNR plot shows this same tradeoff from an error perspective. FPR is the fraction of safe situations in which the system incorrectly issues a warning. FNR is the fraction of actual conflicts the system misses. Conservative thresholds achieve very low FPR (0.010–0.015, rarely warning when unnecessary) but high FNR (0.80–0.85, missing most conflicts). Aggressive thresholds reverse this relationship as they detect more conflicts but warn more often when unnecessary. The $F1$-score plots show that balanced overall performance occurs at intermediate settings. $F1$-score balances precision and recall into a single number. The plots indicate that $F1$ peaks when recall is within 0.4–0.6, and precision is within 0.4–0.5, suggesting these represent balanced operating points. The remaining plots show mathematical relationships between metrics. FPR versus Specificity shows an inverse line because specificity = 1 – FPR by definition. FNR versus recall shows a similar inverse relationship. The Youden's J versus balanced accuracy plot shows these two composite metrics correlate strongly, both capturing overall system balance.

The key insight from Figs. 9 and 10 is that performance changes smoothly and predictably as parameters vary. The smooth color gradients show no abrupt jumps or instabilities.

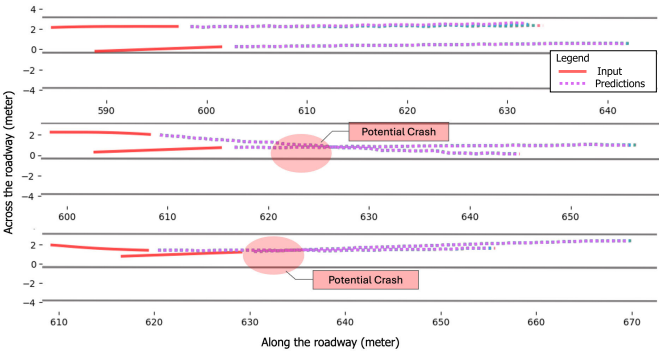


Fig. 11. Predicted trajectories (PT) indicating potential conflict.

Points from smooth curves indicate that the system behaves consistently across different configurations. The baseline configuration ($\lambda = 7, \tau = 0.7$) sits in the conservative region, prioritizing low false alarm rates over maximum detection sensitivity. This demonstrates that the conflict detection framework supports a range of operating modes, from conservative high-confidence warnings to more sensitive advisory alerts, while maintaining stable and predictable behavior throughout.

Fig. 11 presents a potential conflict scenario where one vehicle performs an aggressive lane change maneuver. The visualization of the PT shows a potential conflict before it takes place. The PTs are processed by the warning-generation module and are correctly flagged as a potential conflict. Early detection of conflict by the warning-generation module, using the PT, is the basis for proactive safety, enabling the system to anticipate safety-critical events in advance. The use of HPNet produces temporally consistent and spatially aware

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predictions, thereby enhancing the reliability of these warnings in both constrained and high-risk work zone environments.

VII. Conclusion

This study presents a proactive safety framework for roadway work zones that leverages a PDT architecture enhanced by real-time sensing, HD maps and temporally consistent trajectory prediction using the HPNet model. There are two contributions of this study: 1) development of an infrastructure-enabled vehicle trajectory prediction framework that leverages the HPNet model integrated with Lanelet2 HD maps and 2) development of a proactive safety warning generation application that incorporates probabilistic conflict modeling and vehicle bounding-box representations. By adopting HPNet for the first time in an infrastructure-based deployment, we demonstrate its effectiveness in forecasting vehicle trajectories with high accuracy in a work zone. The cosimulation framework integrating SUMO and CARLA enables realistic data generation and testing, while the use of Lanelet2 maps provides detailed topological encoding, essential for trajectory forecasting in complex work zone environments. In addition, our proactive safety warning generation application demonstrates its ability to issue alerts for potential vehicle conflicts.

Our analyses revealed that the infrastructure-enabled HPNet model demonstrates superior performance on the work zone datasets generated from the cosimulation environment, achieving a minJoint FDE of 0.3228 m and a minJoint ADE of 0.1327 m, lower than the benchmarks on the Argoverse (minJointFDE: 1.0986 m, minJointADE: 0.7612 m) and INTERACTION (minJointFDE: 0.8231 m, minJointADE: 0.2548 m) datasets. For conflict detection, the warning-generation module achieves a low FPR of 1.4% with high specificity of 0.986. These findings underscore the potential to develop PDT for dynamic traffic environments by leveraging infrastructure-based sensing of the physical twin.

While the results indicate the efficacy of the proactive safety warning system, this study is limited by the scope of the test scenarios and the controlled simulated environment in which the evaluation was conducted. Future work will extend this framework to real-world deployments and incorporate cooperative behavior modeling via vehicle-to-everything communication to generate personalized safety warnings, further enhancing safety outcomes.

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